Transition Analysis Toolkit
A Reproducible Analysis of Student Transition Pathways

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3.1 Model Performance

Table 1: ROC AUC comparison for cross-validated Decision Tree and Random Forest models.

Model	CV ROC AUC	Test ROC AUC	Test Rows
Decision Tree — With Risk	0.7072	0.7664	295
Random Forest — Without Risk	0.7101	0.7603	295
Random Forest — With Risk	0.7095	0.7599	295
Decision Tree — Without Risk	0.6994	0.7557	295

3.2 Feature Importance

Rank	Research theme	Mean importance
1	Secondary mathematics preparation	0.2755
2	Composite risk measure	0.2381
3	Preparatory mathematics	0.2320
4	Previous university performance	0.1090
5	Secondary school achievement	0.1058
6	Student characteristics	0.0395

3.3 Feature Rankings

Rank	Feature	Models	Mean importance
1	Composite risk index	2	0.4762
2	Prior preparatory mathematics	4	0.1694
3	Secondary mathematics: Other	4	0.1019
4	Secondary mathematics: Standard	4	0.0950
5	Previous attempt at current subject	4	0.0693
6	Previous preparatory mathematics pass	4	0.0532
7	Previous failure in current subject	4	0.0397
8	International student	4	0.0395
9	Secondary school mark below 70	4	0.0394
10	Secondary mathematics: Extension 1	4	0.0348
11	Secondary school mark 90 or above	4	0.0288
12	Secondary mathematics: Extension 2	4	0.0257
13	Secondary school mark 80–89	4	0.0228
14	Secondary mathematics: Advanced	4	0.0182
15	Secondary school mark 70–79	4	0.0148
16	Previous preparatory mathematics failure	4	0.0094

3.4 Model Interpretation

Across the tuned tree-based models, secondary mathematics preparation emerged as the most influential predictor group (mean importance = 0.276). This was followed by the composite risk measure (0.238) and preparatory mathematics (0.232). Together, these three groups accounted for 74.6% of the average feature importance across the tuned models. Secondary mathematics preparation remained influential across all four models, whereas the composite risk measure contributed strongly but appeared in two of the four models. In contrast, student characteristics made the smallest average contribution. This suggests that prior secondary mathematics preparation was the strongest and most consistent predictor of success across the tuned models.

4 Discussion

5 Conclusion

References

A Supplementary Material